

get the maximum number of operations the processor can perform and the final score is represented as MIPS (millions of instructions per second).

The Whetstone benchmark is used to indicate the FPU (floating point unit) strength. The benchmark consists of a series of floating point operation samples that are run over and over to find the maximum floating point operations the processor can perform—the score is represented as MFLOPS (Million Floating point Operations Per Second). But these MIPS and MFLOPS are only indicative of the kind of performance one can expect from each of the processors.

The best motherboards for content creation

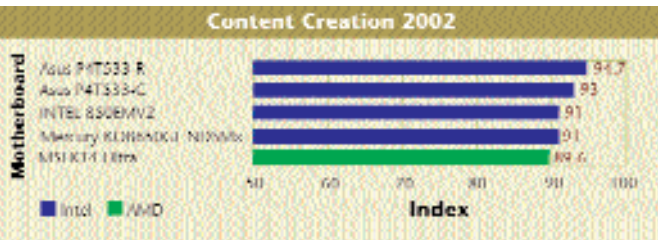
When it comes to productivity, motherboards for the Intel processor dominated the above 70 STTI list. These boards offer excellent performance and provide system stability—a plus for productivity applications that are extremely resource hungry.

However, the highest scorer was an AMD board, the MSI KT4-Ultra with 78.35 STTI. With support for Serial ATA (and the hard disk hot-swapping that this standard brings), this motherboard seems especially great for productivity application areas, where huge amounts of data are routinely transferred between colleagues and machines. Add to that the support for onboard RAID and the reason to invest in a SCSI disk diminishes even further. Bluetooth support will be a boon to those who demand wireless connectivity between cellphones, the PC and peripherals such as printers.

The ASUS A7S266 was the only other AMD solution that crossed the 70 STTI barrier: based on the SiS 740 chipset, the ASUS comes across poorly in supported features but costs about Rs 2,000 less than its MSI counterpart.

Of the eight boards that scored an STTI of more than 70, six belonged to the Intel camp. Chief scorer among them was the Krypton P4TDQ with 77.15 STTI. Based on the 845GL chipset, this motherboard has onboard video, sound, USB 2.0 and Ethernet. Although not exactly rich in features, it nevertheless belts enough of them to be considered a good buy for productivity applications and its price of Rs 3,630 makes it all the more attractive.

Two MSIs and an Intel board ran neck-and-neck with each



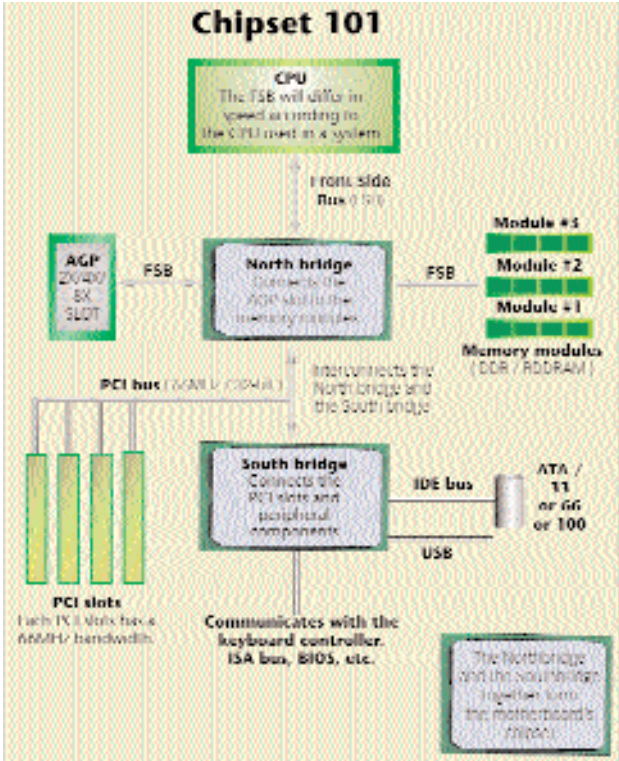
other—the MSI 845PE-Max2, the Intel D845GEBV2 and the MSI 845E-Max2 scored 76.74, 75.62 and 75.52, respectively. The board that stole the thunder though was the Krypton KOB650GL. With a score of 74.86 and a price of Rs 3,600, this SiS chipset-based board came across as a great bargain in this particular application category.

The best processors for content creation

The STTI score sheet for the area of content creation was dominated by AMD processors. Of the nine toppers, only two Intel processors managed to score more than 70 STTI points in this

A Primer on the Chipset

The chipset is the hub of all data transfers and all the components communicate through it. Each chipset consists of a pair of chips: a Northbridge that interfaces with the AGP bus, the processor and the memory bus; and the Southbridge that oversees data transfer between the PCI devices, the disk drives, the network, the USB ports, audio, keyboard controller, etc.



A chipset defines the feature set and the performance of a motherboard and by extension, the entire system. The maximum speed of a processor that the chipset supports is important—for further upgrades only the processor need be changed. For example, if a chipset supports up to 2.8 GHz CPUs and you currently have a 2 GHz CPU, you could upgrade to a 2.8 GHz CPU over a period of time, without having to invest in a new motherboard.

The type of memory that a chipset supports is also important. There are three types of memory running at different speeds—SDRAM (the slowest type), DDR SDRAM (faster than SDRAM) and RDRAM (theoretically, the fastest). The DDR variant is all set to become the new standard for memory as its prices are falling and RDRAM is being increasingly marginalised.

category.

The AMD processors, when first introduced, sported a 200 MHz FSB. As technology improved, we were able to get our hands on Athlons running on a 266 MHz system bus. The fastest AMD processors now operate at 333 MHz FSB and the first ones to extend support for this speed increase are the Thoroughbred range of processors from AMD—the 2700+ and the 2800+. Currently the chipsets that offer support for the DDR333 standard are nVidia's nForce2 and VIA's KT400 and KT333. The 333 MHz bus theoretically delivers a bus speed of